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The Prevalence and Influence of New or Worsened Neck Pain After a Sport-Related Concussion in Collegiate Athletes

A Study From the CARE Consortium

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Background: Neck pain in a concussion population is an emerging area of study that has been shown to have a negative influence on recovery. This effect has not yet been studied in collegiate athletes.

Hypothesis: New or worsened neck pain is common after a concussion (>30%), negatively influences recovery, and is associated with patient sex and level of contact in sport.

Study Design: Cohort study; Level of evidence, 2.

Methods: Varsity-level athletes from 29 National Collegiate Athletic Association member institutions as well as nonvarsity sport athletes at military service academies were eligible for enrollment. Participants completed a preseason baseline assessment and follow-up assessments at 6 and 24 to 48 hours after a concussion, when they were symptom-free, and when they returned to unrestricted play. Data collection occurred between January 2014 and September 2018.

Results: A total of 2163 injuries were studied. New or worsened neck pain was reported with 47.0% of injuries. New or worsened neck pain was associated with patient sex (higher in female athletes), an altered mental status after the injury, the mechanism of injury, and what the athlete collided with. The presence of new/worsened neck pain was associated with delayed recovery. Those with new or worsened neck pain had 11.1 days of symptoms versus 8.8 days in those without ($P < .001$). They were also less likely to have a resolution of self-reported symptoms in $\leq$7 days ($P < .001$). However, the mean duration of the return-to-play protocol was not significantly different for those with new or worsened neck pain (7.5 $\pm$ 7.7 days) than those without (7.4 $\pm$ 8.3 days) ($P = .592$).

Conclusion: This novel study shows that neck pain was common in collegiate athletes sustaining a concussion, was influenced by many factors, and negatively affected recovery.

Keywords: concussion; neck pain; whiplash; sports medicine

Concussions are commonly related to sports or recreational activities in the United States.[6] As many as 3.8 million sport/recreation-related concussions occur in the United States each year.[21]

In the past decade, there has been increased attention toward the possibility of comorbid cervical spine (ie, whiplash) injuries and their protentional effect on recovery from a concussion.[†] The prevalence of neck pain after a sport-related concussion (SRC) has been estimated to range between 34% and 54% in pediatric populations.[8,33] The cervical spine has frequently been acknowledged as a source of persisting concussion-related symptoms,[12] with evidence suggesting that patients who report neck pain after a concussion have an increased risk of developing persistent postconcussion symptoms.[9,12,32,33] To date,

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however, the majority of these investigations have been largely focused on non-SRCs or pediatric SRCs only.

Recent studies and case series have begun to show that treatment involving the cervical spine after a concussion has the possibility to improve outcomes such as a reduction in persistent postconcussion symptoms and a decrease in the time to return to activity and play.[17,18,25,36] Despite the growth in attention being paid to concurrent neck pain and concussions, there still exists a knowledge gap of baseline and injury characteristics that influence whether an athlete is at a greater risk for new or worsened neck pain after a concussion. Further, the effect of neck pain on recovery in a collegiate population that has suffered an SRC is unknown. Therefore, the objectives of this study were to (1) estimate the frequency of new/worsened neck pain after a concussion in comparison with baseline in a population of collegiate athletes, (2) identify both baseline and injury variables that influence whether an athlete will develop new/worsened neck pain after an SRC, and (3) understand how the presence of new/worsened neck pain affects the length of recovery after a concussion. On the basis of previous literature, we hypothesized that new or worsened neck pain would be a common finding after a concussion and that the presence of new or worsened neck pain would be influenced by level of contact in sport, sex (ie, greater in female athletes), and mechanism of injury in which the head struck the playing surface.[8,20,33,37,38] We also hypothesized that new or worsened neck pain would negatively affect the length of recovery after a concussion.[8,33]

METHODS

This study is a prospective cohort study using a secondary analysis of the Concussion Assessment, Research and Education (CARE) Consortium sample. All male and female varsity-level athletes from 29 National Collegiate Athletic Association (NCAA) member institutions across the United States as well as all male and female nonvarsity sport athletes at military service academies were eligible for enrollment. Data collection occurred between January 2014 and September 2018. All participants completed a preseason baseline assessment that included demographic characteristics, medical history, balance and neurocognitive testing, and self-reported symptoms. For the purposes of this study, a definition of a concussion developed through evidence-based guidelines was adopted. The CARE Consortium adopted the definition of a concussion as "a change in brain function after a force to the head, which may be accompanied by temporary loss of consciousness, but is identified in awake individuals with measures of neurologic and cognitive dysfunction."[4] For this study, and consistent with conventions in the clinical care of athletes with concussions, the presence of concussion symptoms

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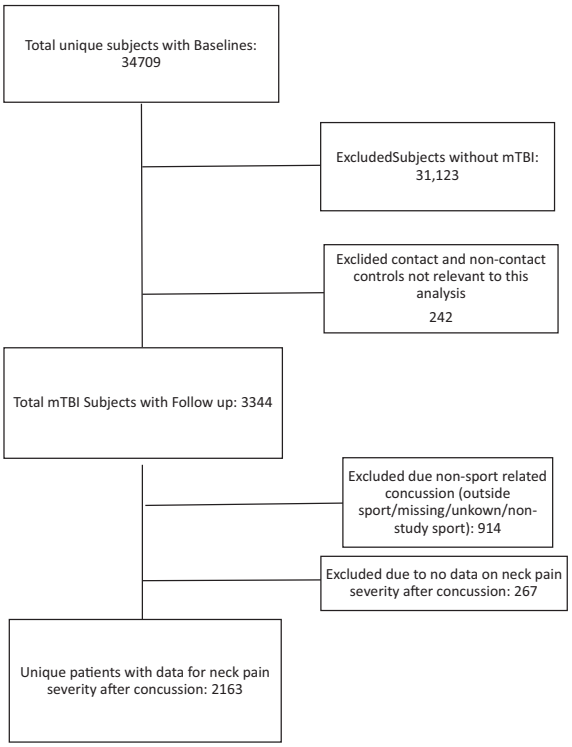

Figure 1. CONSORT (Consolidated Standards of Reporting Trials) diagram.

was included as an indication of a change in brain function. After a concussion was suspected, athletes were evaluated by a clinician/sports medicine professional, and the determination of a concussion was based on the above definition. They then completed follow-up assessments at 6 and 24 to 48 hours after the injury. The athlete was assessed again when symptom-free and then when cleared for unrestricted return to play. The institutional review boards of record operate in accordance with the standards of ethics outlined in the Declaration of Helsinki. The CARE Consortium's design and methods, including the institutional review board approval process, are described in detail elsewhere.[3] A CONSORT (Consolidated Standards of Reporting Trials) diagram for the sample included in this study is provided in Figure 1.

Data Collection

After enrollment, demographic characteristics, patient and family health history, and current and previous sport history were collected. Demographic variables included weight, height, body mass index, age, sex, and race and ethnicity. Sport-related variables included primary sport and years spent playing the primary sport. At baseline, the outcome measures of interest were also administered, which included the Standardized Assessment of Concussion (SAC) to evaluate cognitive functioning,[2,26,28] the Balance Error Scoring System (BESS) to evaluate postural stability,[14,27] and 2 measures of self-reported symptoms including the Sport Concussion Assessment Tool 3 (SCAT3) symptom checklist and the Brief Symptom Inventory–18 (BSI-18) to evaluate depression, anxiety, and somatization.[29] These outcome measures were administered at all time points. The primary outcome of interest was new or worsened neck pain derived from baseline and postinjury ratings of neck pain on the SCAT3 (from 0 [none] to 6 [severe] at each time point).

The characteristics of the injury included the following: injury situation (practice/training or competition), mechanism of injury, who/what the injured athlete collided with, impact location resulting in the injury, whether there was a delayed onset of symptoms, whether the athlete had an alteration in mental status (eg, dazed, stunned, confused, saw stars), whether the athlete lost consciousness, whether the athlete had posttraumatic amnesia, and whether the athlete had retrograde amnesia. The mechanism of injury included contact with the ball/puck/stick, soccer header with physical contact, soccer header without physical contact, multiple, other, and player-to-player contact. What the injured athlete collided with described what object or body part the player specifically collided with, including equipment, multiple, other player's body, other player's head/helmet, turf, ground, floor, ice, and other.

Acute postinjury measures included the following: SCAT3 total symptom severity score (0-132), total number of symptoms on the SCAT3 (0-22), SAC total score (0-30), BESS total score (0-60), and BSI-18 global severity index score (0-72). Recovery was documented in multiple ways including the number of days in which the athlete experienced concussion-related symptoms, the number of days before the athlete returned to normal academic performance (ie, felt normal at school without restrictions) as reported by the athlete, the duration of the return-to-play protocol, and the duration of self-reported symptoms. The duration of self-reported symptoms was categorized as <24 hours, 24 hours to 7 days, 8 days to 1 month, 1-6 months, or 6 months after the injury.

Statistical Analysis

The primary outcome was new or worsened neck pain reported at either 6 or 24 to 48 hours after a concussion on the SCAT3, defined as at least a 1-point increase in the 0-to-6 rating of neck pain from baseline to the postinjury time point.

Descriptive statistics obtained included counts (percentages) for categorical variables and the median (range) or mean ± SD for continuous variables. Bivariate associations with new or worsened neck pain were assessed using the chi-square test for categorical variables and the Wilcoxon rank-sum test for continuous variables. An "unknown" category was created for missing values of categorical variables. Missingness was controlled for using complete case analysis.

Self-reported race and ethnicity were combined into 1 race/ethnicity variable, with categories for Hispanic, White, Black, other, and unknown.

To assess the relationship between sport and new or worsened neck pain, sports were classified by the level of contact (collision, contact, limited contact, and noncontact)

as previously described in the literature.[35] This was first evaluated using the chi-square test and later post hoc analysis comparing specific sports.

Multivariable logistic regression was used to assess the association of new or worsened neck pain with baseline and injury variables. This modeling used forward stepwise selection with all the variables listed above as candidates. Specifically, an effect not yet in the model was included if it had the lowest P value not exceeding .30 among all the effects not yet in the model, while an effect already in the model was removed if, after the addition of a new variable, its statistical significance increased above .35. The model selection stopped when all effects in the model were significant at the .35 level and none of the candidates outside the model were significant at the .30 level. The relatively high inclusion/removal significance levels were chosen to reduce the bias associated with the model selection and uncertainty about the inclusion of effects with significance close to .05. Odds ratios with 95% CIs were calculated for pairwise comparisons of all levels of the categorical variables included in the final model. In the logistic regression model, the P values of the pairwise comparisons of categories were adjusted for multiple testing using the Tukey-Kramer method.

The following variables were included in the model selection: level of contact in sport, weight, height, body mass index, age, sex, previous concussions, race/ethnicity, sleep disorder, balance disorder, psychiatric disorder, learning disorder, attention deficit-hyperactivity disorder, depression, moderate/severe traumatic brain injury, vision problems, hearing problems, Parkinson disease, memory disorder, migraines, headache disorder (nonmigraine), general anesthesia, number of instances of general anesthesia, mean hours of sleep each night for the past week (Sunday-Thursday), mean hours of sleep each night for the past weekend (Friday and Saturday), injury situation, primary position (for football only), mechanism of injury, what the injured athlete collided with, impact location resulting in the injury, whether there was a delayed onset of symptoms, whether the athlete had an alteration in mental status, whether the athlete lost consciousness, whether the athlete had posttraumatic amnesia, whether the athlete had retrograde amnesia, baseline SCAT3 total symptom severity score, baseline total number of symptoms on the SCAT3, baseline SAC total score, baseline BESS total score, baseline BSI-18 global severity index score, and 3 subscale scores of the BSI-18.

Further analyses of the mechanism of injury and what the injured athlete collided with were completed by performing a series of post hoc tests comparing each possible combination of variables for each category.

The time course of symptom recovery was categorized as <24 hours, 24 hours to 7 days, 8 days to 1 month, 1-6 months, or 6 months. The effect of new or worsened neck pain acutely after the injury on recovery variables was evaluated using the Wilcoxon rank-sum test. To determine which categories of self-reported recovery were most affected, a series of post hoc tests separating the ordinal symptom duration outcome into 2 groups with all possible dichotomizations was used.

RESULTS

Participant and Injury Characteristics

The demographic characteristics and baseline outcomes of the sample are presented in Table 1. The final sample included 2163 athletes with a concussion. There were 1016 (47.0%) athletes who reported new or worsened neck pain after their concussion. The characteristics of the injury leading to a concussion diagnosis are detailed in Table 2.

Factors Associated With New or Worsened Neck Pain

Stepwise logistic regression revealed several factors that were associated with new or worsened neck pain after a concussion. These factors included female sex, a higher mean number of hours of sleep each night at baseline, a higher level of contact in sport, the mechanism of injury, what the athlete collided with, and an altered mental status. Table 3 details the post hoc pairwise comparisons for all factors associated with new or worsened neck pain after a concussion with stepwise logistic regression.

A secondary analysis examined the relationship between sex and neck pain including only sports with 20 injuries to both male and female athletes, as well as for each sex, which revealed female athletes to be more likely to report new or worsened neck pain (odds ratio, 1.586; P = .0284).

Effect of New or Worsened Neck Pain on Recovery

Athletes with new or worsened neck pain acutely after the injury were significantly less likely to have a resolution of self-reported symptoms of $\leq$7 days, were more likely to have a larger number of symptomatic days, and take longer to return to normal academic performance. However, the mean length of the return-to-play protocol, which was started once asymptomatic, was not significantly different for those with new or worsened neck pain than for those without. A detailed presentation of these findings can be found in Table 4.

DISCUSSION

To our knowledge, this is the first study to date to assess the prevalence of neck pain in collegiate athletes diagnosed with an SRC. In addition, this is the largest study, to our knowledge, to examine the association between acute postconcussive neck pain and concussion recovery. In 2163 prospectively recruited collegiate athletes with a concussion, 47.0% reported new-onset or worsened neck pain after a concussion, and multiple preinjury and injury-related factors influenced the manifestation of new or worsened neck pain. New or worsened neck pain acutely after the injury was associated with a longer recovery duration after the SRC. Findings highlight the importance of evaluating and addressing the presence of neck pain as part of a comprehensive approach to managing SRCs.

TABLE 1
Demographic Characteristics and Baseline Outcomes of Athletes[a]

	Total (N = 2163)	No New/Worsened Neck Pain (n = 1147)	New/Worsened Neck Pain (n = 1016)	*P*
Sex				.117[b]
Female	793 (36.7)	403 (50.8)	390 (49.2)	
Male	1370 (63.3)	744 (54.3)	626 (45.7)	
Race/ethnicity				.725[b]
Hispanic	157 (7.3)	91 (58.0)	66 (42.0)	
Black	413 (19.1)	212 (51.3)	201 (48.7)	
Other	250 (11.6)	134 (53.6)	116 (46.4)	
White	1328 (61.4)	702 (52.9)	626 (47.1)	
Unknown	15 (0.7)	8 (53.3)	7 (46.7)	
Age, y				.745[c]
Median (range)	19.0 (17.0-25.0)	19.0 (17.0-25.0)	19.0 (17.0-24.0)	
Mean ± SD	19.2 ± 1.3	19.3 ± 1.4	19.2 ± 1.3	
Missing, n	0	0	0	
Mean hours of sleep each night (Sunday-Thursday)				.195[c]
Median (range)	7.0 (2.0-24.0)	7.0 (2.0-24.0)	7.0 (3.0-24.0)	
Mean ± SD	7.1 ± 2.0	7.2 ± 2.3	7.0 ± 1.6	
Missing, n	55	36	19	
SCAT3 total symptom severity score				.817[c]
Median (range)	2.0 (0.0-103.0)	2.0 (0.0-103.0)	2.0 (0.0-62.0)	
Mean ± SD	5.2 ± 8.9	5.4 ± 9.6	5.0 ± 8.2	
Missing, n	0	0	0	
SCAT3 total number of symptoms				.779[c]
Median (range)	1.0 (0.0-22.0)	1.0 (0.0-22.0)	1.0 (0.0-21.0)	
Mean ± SD	2.8 ± 3.8	2.8 ± 3.9	2.8 ± 3.8	
Missing, n	0	0	0	
SAC total score				.491[c]
Median (range)	28.0 (19.0-30.0)	28.0 (19.0-30.0)	28.0 (19.0-30.0)	
Mean ± SD	27.4 ± 2.0	27.4 ± 2.0	27.3 ± 2.1	
Missing, n	50	28	22	
BESS total score				.192[c]
Median (range)	12.0 (1.0-41.0)	13.0 (1.0-41.0)	12.0 (1.0-40.0)	
Mean ± SD	13.4 ± 6.5	13.6 ± 6.7	13.1 ± 6.2	
Missing, n	88	46	42	
BSI-18 global severity index score				.888[c]
Median (range)	0.0 (0.0-48.0)	0.0 (0.0-46.0)	0.0 (0.0-48.0)	
Mean ± SD	2.6 ± 5.0	2.6 ± 5.0	2.6 ± 5.1	
Missing, n	29	21	8	

[a]Data are shown as n (%) unless otherwise indicated. BESS, Balance Error Scoring System; BSI-18, Brief Symptom Inventory–18; SAC, Standardized Assessment of Concussion; SCAT3, Sport Concussion Assessment Tool 3.
[b]Chi-square test.
[c]Wilcoxon rank-sum test.

The prevalence of neck pain in our sample is similar to other reports of neck pain in SRC studies (34%-54%).[8,12,33] However, in contrast to these previous studies, we had the benefit of access to baseline data, which allowed us to evaluate new or worsened neck pain. This allows for athlete-specific considerations and comparisons with preinjury levels of preexisting neck pain. To our knowledge, previous studies have not reported on any preinjury levels of neck pain.

In contrast to a previous study, we found that contact with the playing surface was associated with higher odds of neck pain in comparison to contact with another player.[8] The previous article was a study of high school athletes who were less developmentally and physically mature and were likely to have less neck strength than the current study's sample of Division I collegiate athletes. Greater neck strength may decrease linear acceleration of the head, related to contact during sport, thus making contact with a player less likely to produce neck pain.[10,11]

Regarding the level of contact in sport, positive odds ratios were found when comparing categories of high contact to lower contact in every comparison, with the exception of contact to limited contact sports. More frequent contact provides greater opportunity for forces capable of causing concurrent cervical spine injuries. However, because the prevalence of new/worsened neck pain was not determined in athletes who were not diagnosed with a concussion, we cannot conclude that this difference in rates of new or worsened neck pain is different than those experienced by athletes without concussions.

TABLE 2
Injury Characteristics Related to Concussion Diagnosis[a]

	Total (N = 2163)	No New/Worsened Neck Pain (n = 1147)	New/Worsened Neck Pain (n = 1016)	P
Level of contact				.058[b]
Collision	812 (37.5)	400 (49.3)	412 (50.7)	
Contact	719 (33.2)	405 (56.3)	314 (43.7)	
Limited contact	175 (8.1)	89 (50.9)	86 (49.1)	
Noncontact	61 (2.8)	33 (54.1)	28 (45.9)	
Unassigned	396 (18.3)	220 (55.6)	176 (44.4)	
Injury situation				.005[b]
Competition	734 (33.9)	420 (57.2)	314 (42.8)	
Practice/training	1429 (66.1)	727 (50.9)	702 (49.1)	
Occurrence of injury during competitive season				.298[b]
No	295 (21.2)	142 (48.1)	153 (51.9)	
Yes	1096 (78.8)	565 (51.6)	531 (48.4)	
Missing, n	772	440	332	
Mechanism of injury				<.001[b]
Contact with ball/puck/stick	298 (13.9)	179 (60.1)	119 (39.9)	
Header with physical contact	45 (2.1)	31 (68.9)	14 (31.1)	
Header without physical contact	22 (1.0)	18 (81.8)	4 (18.2)	
Multiple	122 (5.7)	60 (49.2)	62 (50.8)	
Other	508 (23.7)	255 (50.2)	253 (49.8)	
Player-to-player contact	1150 (53.6)	596 (51.8)	554 (48.2)	
Missing, n	18	8	10	
Object that injured athlete collided with				<.001[b]
Equipment	68 (3.2)	37 (54.4)	31 (45.6)	
Multiple	119 (5.6)	65 (54.6)	54 (45.4)	
Other	517 (24.4)	289 (55.9)	228 (44.1)	
Other player's body	594 (28.0)	336 (56.6)	258 (43.4)	
Other player's head/helmet	548 (25.8)	290 (52.9)	258 (47.1)	
Turf/ground/floor/ice	274 (12.9)	104 (38.0)	170 (62.0)	
Missing, n	43	26	17	
Alteration in mental status				.014[b]
No	1025 (48.3)	572 (55.8)	453 (44.2)	
Yes	1097 (51.7)	554 (50.5)	543 (49.5)	
Missing, n	41	21	20	
Loss of consciousness				.080[b]
No	2044 (95.4)	1092 (53.4)	952 (46.6)	
Yes	99 (4.6)	44 (44.4)	55 (55.6)	
Missing, n	20	11	9	
Posttraumatic amnesia				.016[b]
No	1926 (90.1)	1040 (54.0)	886 (46.0)	
Yes	212 (9.9)	96 (45.3)	116 (54.7)	
Missing, n	25	11	14	
Retrograde amnesia				.494[b]
No	2023 (94.7)	1078 (53.3)	945 (46.7)	
Yes	114 (5.3)	57 (50.0)	57 (50.0)	
Missing, n	26	12	14	

[a]Data are shown as n (%) unless otherwise indicated.
[b]Chi-square test.

Consistent with the existing literature, our study showed that female athletes were more likely to report neck pain after a concussion.[8,33] A recent systematic review reported higher levels of overall symptom endorsement at baseline and after the injury by female participants.[30] However, our study also uniquely replicated this association in a subsample of sports with similar rules for both male and female athletes. Female athletes may be at a higher risk for cervical injuries, given a decreased ratio of neck mass to head size, allowing for greater transmission of forces to the head.[15]

Regarding recovery after a concussion, findings showed a longer time needed to return to normal academic performance and a greater number of symptomatic days, which were all self-reported variables. Once the athlete was asymptomatic, the return-to-play protocol, which represents the time from becoming asymptomatic to being cleared to full participation in sport, was initiated, but

TABLE 3
Factors Associated With New or Worsened Neck Pain

	Odds Ratio (95% CI)	*P*
Mean hours of sleep each night (Sunday-Thursday)		
Per 1-hour increase	0.94 (0.90-0.99)	.023
Alteration in mental status		
Yes vs no	1.28 (1.05-1.56)	.016
Sex		
Female vs male	1.48 (1.17-1.89)	.001
Level of contact		.004
Collision vs contact	1.42 (1.12-1.81)	.006
Collision vs limited contact	1.01 (0.68-1.51)	.956
Collision vs noncontact	1.61 (0.91-2.85)	.103
Collision vs unassigned	1.46 (1.10-1.95)	.010
Contact vs limited contact	0.71 (0.49-1.03)	.075
Contact vs noncontact	1.14 (0.65-1.97)	.650
Contact vs unassigned	1.03 (0.79-1.35)	.809
Limited contact vs noncontact	1.59 (0.86-2.96)	.142
Limited contact vs unassigned	1.45 (0.97-2.17)	.074
Mechanism of injury		.007
Contact with ball/puck/stick vs header with physical contact	1.00 (0.47-2.13)	.994
Contact with ball/puck/stick vs header without physical contact	2.40 (0.77-7.44)	.130
Contact with ball/puck/stick vs multiple	0.49 (0.29-0.82)	.007
Contact with ball/puck/stick vs other	0.57 (0.40-0.83)	.003
Player-to-player contact vs contact with ball/puck/stick	1.88 (1.24-2.84)	.003
Player-to-player contact vs header with physical contact	1.89 (0.96-3.69)	.064
Player-to-player contact vs header without physical contact	4.51 (1.44-14.12)	.010
Player-to-player contact vs multiple	0.91 (0.62-1.34)	.6382
Player-to-player contact vs other	1.08 (0.81-1.45)	.581
Header with physical contact vs header without physical contact	2.39 (0.65-8.75)	.188
Header with physical contact vs other	0.58 (0.29-1.16)	.124
Header without physical contact vs other	0.24 (0.08-0.74)	.013
Multiple vs other	1.19 (0.84-1.77)	.010
Object that injured athlete collided with		.001
Equipment vs multiple	1.35 (0.70-2.58)	.370
Equipment vs other	1.04 (0.62-1.74)	.883
Equipment vs other player's body	1.41 (0.80-2.49)	.237
Equipment vs other player's head/helmet	1.29 (0.72-2.32)	.396
Turf/floor/ground/ice vs equipment	1.51 (0.85-2.67)	.162
Turf/floor/ground/ice vs multiple	2.03 (1.28-3.21)	.003
Turf/floor/ground/ice vs other	1.57 (1.10-2.22)	.012
Turf/floor/ground/ice vs other player's body	2.13 (1.54-2.93)	<.0001
Turf/floor/ground/ice vs other player's head/helmet	1.94 (1.38-2.73)	.0001

there was no difference in the length of the return-to-play protocol.

Our study builds on previous studies and indicates that athletes with neck pain after a concussion should undergo a specific cervical spine examination to assess for concurrent cervical spine and/or whiplash-related disorders. Given the likelihood of a greater number of days with concussion symptoms and a longer period of time with deleterious academic effects in those with concurrent concussions, these patients should be considered for an early intervention targeted at the cervical spine. Growing evidence from case reports, case series, and 1 small trial suggests that targeted cervical rehabilitation could benefit this population once persistent symptoms have already developed.[13,17,18,25,36] Further research including randomized controlled trials is needed to better understand the potential benefit of specific cervical spine interventions in this population with the goal of shortening recovery in patients with new or worsened neck pain after a concussion.

Limitations

Limitations of this study included missing neck pain severity scores in a subsample (n = 267) and a lack of objective assessments of the neck to confirm a concurrent cervical spine injury. New neck pain was also not separately assessed in comparison with worsened neck pain. Additionally, this sample was restricted to NCAA athletes as well as military service academy members and, as such, may not be generalizable to other levels of play or non-SRCs. However, several recent publications in the civilian population have also found neck pain to be both common and predictive of delayed recovery after a concussion.[19,20,22]

TABLE 4
Effect of New or Worsened Neck Pain on Recovery[a]

	Total (N = 2163)	No New/Worsened Neck Pain (n = 1147)	New/Worsened Neck Pain (n = 1016)	*P*
Duration of self-reported symptoms, d				<.001[b]
Median (range)	6.9 (0.2-123.9)	5.9 (0.2-123.9)	8.0 (0.6-103.7)	
Mean ± SD	9.9 ± 11.0	8.8 ± 10.8	11.1 ± 11.1	
Missing, n	240	130	110	
Duration of return-to-play protocol, d				.592[b]
Median (range)	5.0 (0.0-131.0)	5.0 (1.0-131.0)	5.0 (0.0-99.0)	
Mean ± SD	7.5 ± 8.0	7.4 ± 8.3	7.5 ± 7.7	
Missing, n	527	270	257	
Time to resolution of self-reported symptoms				<.001[b]
<24 hours	76 (5.8)	53 (7.7)	23 (3.7)	
1-7 days	636 (48.7)	368 (53.7)	268 (43.2)	
8-30 days	517 (39.6)	230 (33.6)	287 (46.2)	
>1 month	68 (5.2)	31 (4.5)	37 (6.0)	
6 months	9 (0.7)	3 (0.4)	6 (1.0)	
Missing, n	857	462	395	
Time to resolution of self-reported symptoms (secondary analysis)				.002[c]
<24 hours	76 (5.8)	53 (7.7)	23 (3.7)	
1-7 days/8-30 days/>1 month/6 months	1230 (94.2)	632 (92.3)	598 (96.3)	
Missing, n	857	462	395	
Time to resolution of self-reported symptoms (secondary analysis)				<.001[c]
<24 hours/1-7 days	712 (54.5)	421 (61.5)	291 (46.9)	
8-30 days/>1 month/6 months	594 (45.5)	264 (38.5)	330 (53.1)	
Missing, n	857	462	395	
Time to return to normal academic performance, d				<.001[b]
Median (range)	2.0 (0.0-96.0)	2.0 (0.0-96.0)	3.0 (0.0-72.0)	
Mean ± SD	5.2 ± 8.3	4.7 ± 8.2	5.7 ± 8.4	
Missing, n	478	237	241	

[a]Data are shown as n (%) unless otherwise indicated.
[b]Wilcoxon rank-sum test.
[c]Chi-square test.

CONCLUSION

This study shows that new or worsened neck pain was common in this population, being reported by 47.0% of injured athletes, and was influenced by many factors including sex, altered mental status after the injury, and what the athlete collided with. Regarding recovery, new or worsened neck pain was associated with an increased total number of days with symptoms and a longer time to return to normal academic performance. This suggests that athletes with neck pain after a concussion should undergo a cervical spine examination. Further research including randomized controlled trials is needed to better understand the potential benefit of specific cervical spine interventions in this population with the goal of shortening recovery in patients with new or worsened neck pain after a concussion.

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